

CALIBRATED HIERARCHICAL VALUE ALIGNMENT FOR LARGE LANGUAGE MODELS

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ABSTRACT

Aligning and evaluating large language models (LLMs) across heterogeneous human value systems is fundamentally hindered by uncalibrated intensity scales, boundary saturation in preference evaluation, and dimensional entanglement in value representations. In this work, we propose Calibrated Hierarchical Value Alignment (CHVA), a unified framework comprising three synergistic components: (1) Hierarchical Contrastive Projection (HCP) with a mathematically corrected off-diagonal orthogonality loss to disentangle taxonomy-aware value embeddings; (2) Per-Value Calibration (PVC) via baseline mean-centering and variance standardization (z -scoring) to eliminate LLM default endorsement bias; and (3) Linear Non-Saturating Extrapolation (LNE) to continuously evaluate out-of-domain extreme value expressions without boundary saturation. Empirical evaluations show that HCP improves pairwise ranking accuracy (+20.3%) and similarity correlation (+38.5%) over base encoders while reducing cross-value vector overlap from 0.404 to 0.119—the lowest disentanglement error among external hierarchical contrastive baselines (HGCLR, HCL-MTC, DCL), which collapse to near-collinear value axes ($|\cos \theta|$ up to 0.980). On the OpinionQA demographic alignment benchmark, PVC-guided value profiling achieves statistically significant accuracy gains over uncalibrated profiles across multiple models ($p < 10^{-5}$ on `gemini-2.5-flash` and $p < 10^{-6}$ on `gemini-2.5-flash-lite`). Furthermore, LNE restores steering evaluation fidelity ($\rho = 0.717$ vs. 0.706 for Plackett-Luce) under extreme targets (± 2). Finally, our collateral capability analysis uncovers critical, value-asymmetric safety erosion bounds under extreme steering (e.g., safety refusal dropping to 51.7% under negative security steering), establishing concrete operational limits for steerable alignment deployments.

1 INTRODUCTION

As large language models (LLMs) transition from open-ended instruction followers to autonomous social agents, aligning their behavioral profiles with pluralistic human value systems has become a central challenge in machine learning research Yao et al. (2023); Bai et al. (2022). Early alignment methodologies primarily emphasized general helpfulness and harmlessness using normative benchmark scenarios Hendrycks et al. (2020); Reinig et al. (2024). However, real-world deployment requires modeling fine-grained, culturally nuanced principles defined by psychological theories such as Schwartz Value Theory (SVT) and Moral Foundations Theory (MFT) Kiesel et al. (2023); Mirzakhmedova et al. (2023). Despite recent progress in structured value representation and relative preference aggregation Kim et al. (2026); Xu et al. (2025); Nie et al. (2025), current value alignment frameworks suffer from three pervasive methodological bottlenecks:

- 1. Vector Coupling and Dimensional Entanglement:** Unregularized contrastive representation heads suffer from dimensional collapse Jing et al. (2021), causing distinct value axes (e.g., *Benevolence* vs. *Universalism*) to exhibit high cross-vector similarity Park et al. (2023); Braun et al. (2025); Marks & Tegmark (2023), while naive orthogonality regularizers hit zero-gradient failure modes on unit-norm embeddings, leaving directional overlap unmitigated.
- 2. Uncalibrated Intensity and Default Endorsement Bias:** Unnormalized scalar scoring reflects the model’s intrinsic default endorsement biases rather than true demographic distinctiveness Hu & Collier (2024); Santurkar et al. (2023), so high-endorsement values dominate

persona selection, obscuring group-specific priorities and degrading accuracy on underrepresented populations Guo et al. (2025).

3. **Boundary Saturation in Intensity Evaluation:** Standard preference judges (e.g., Plackett-Luce) clamp estimated utilities to reference anchor bounds Zheng et al. (2023); Gao et al. (2022); under extreme steering targets (± 2), these saturate at artificial ceilings, forfeiting rank resolution and violating ordinal monotonicity for out-of-domain responses Marro et al. (2026); Robertson (2025).

To overcome these structural limitations, we introduce **Calibrated Hierarchical Value Alignment (CHVA)**, a principled and streamlined framework designed to achieve accurate, disentangled, and non-saturating value alignment across representation, evaluation, and demographic steering. CHVA advances the value alignment paradigm through three core components:

- **Hierarchical Contrastive Projection (HCP):** A multi-level supervised contrastive loss paired with a rigorous off-diagonal orthogonality regularizer $\mathcal{L}_{\text{orth}} = \frac{1}{K(K-1)} \sum_{i \neq j} \langle v_i, v_j \rangle^2$ that—unlike flawed diagonal penalties which vanish on unit vectors—actively reduces cross-value overlap while improving ranking accuracy and correlation.
- **Per-Value Calibration (PVC):** A standardized z -score transform centering per-value scores on global population parameters, eliminating endorsement bias and rendering intensities commensurable across heterogeneous theories.
- **Linear Non-Saturating Extrapolation (LNE):** A smooth single-slope extrapolation extending preference-judge evaluation beyond fixed anchor ranges, preventing boundary saturation at extreme targets (± 2) while preserving ordinal monotonicity.

We conduct extensive empirical evaluations across multiple open-ended value benchmarks. HCP substantially improves ranking accuracy and correlation over base encoders and external hierarchical contrastive baselines while drastically reducing cross-value overlap; PVC-guided profiling on OpinionQA Santurkar et al. (2023) matches state-of-the-art multi-LLM pluralistic alignment at a fraction of the cost; and we establish a full-distribution Pareto frontier mapping steering intensity against collateral degradation on reasoning (GSM8K), factual knowledge (TriviaQA), and safety refusal (AdvBench), uncovering value-asymmetric safety risks that yield actionable operational bounds for steerable deployments.

2 RELATED WORK

Our work intersects three active domains of language model alignment: value representation spaces, steerable demographic alignment, and preference ranking models.

2.1 VALUE REPRESENTATION AND EMBEDDING SPACES

Quantifying human values in text requires mapping open-ended language to structured psychological frameworks, such as Schwartz Value Theory (SVT) and Moral Foundations Theory (MFT). Benchmark corpora including Social Chemistry Forbes et al. (2020), ValueNet Qiu et al. (2021), the Moral Foundations Reddit Corpus Trager et al. (2022), and ValueEval Mirzakhmedova et al. (2023) have laid the foundation for training value-aware models. Modern text embedding backbones, such as the Qwen3 embedding series Yang et al. (2025); Zhang et al. (2025b), leverage Matryoshka Representation Learning (MRL) Kusupati et al. (2022) to compress semantic information into fixed-dimensional vectors. Unified spaces like UniVar Cahyawijaya et al. (2024) and VALUEFLOW’s HIVES Kim et al. (2026) map diverse corpora into continuous value coordinates.

However, standard embedding objectives often fail to maintain taxonomic hierarchy and directional independence. Methods incorporating hierarchy, such as Hierarchy-guided Contrastive Learning (HGCLR) Wang et al. (2022) and Hierarchical Contrastive Learning for Multi-label Text Classification (HCL-MTC) Zhang et al. (2025a), pull ancestor-sharing concepts together but do not explicitly enforce vector orthogonality across distinct value dimensions. Consequently, representations suffer from dimensional collapse Jing et al. (2021), causing cross-value semantic entanglement. While Decoupled Contrastive Learning (DCL) Yeh et al. (2021) removes positive-negative coupling in InfoNCE, it does not prevent directional overlap among macro-value axes. A complementary self-

supervised line counteracts collapse via explicit *off-diagonal* decorrelation—Barlow Twins Zbontar et al. (2021) penalizes off-diagonal cross-correlation across batch feature dimensions and VICReg Bardes et al. (2021) adds variance/covariance terms. Our Hierarchical Contrastive Projection (HCP) transplants this principle to *taxonomic macro-value direction vectors*, combining multi-level ancestry contrast with a corrected off-diagonal orthogonality penalty that delivers non-zero gradients to enforce mutual orthogonality among value axes.

2.2 LLM VALUE STEERING AND DEMOGRAPHIC ALIGNMENT

Controlling model stance across diverse user communities is vital for pluralistic alignment Sorensen et al. (2024); Kirk et al. (2024). Prior work explores demographic alignment on public opinion benchmarks such as OpinionQA Santurkar et al. (2023). Prominent paradigms include persona prompting Hu & Collier (2024), multi-LLM collaborative architectures like Modular Pluralism Feng et al. (2024), dynamic activation frameworks such as VISPA Zheng et al. (2026), counterfactual structural causal modeling via COUPLE Guo et al. (2025), and survey-to-behavior fine-tuning Nie et al. (2025). Beyond prompt engineering, mechanistic steering techniques rely on representation engineering (RepE) Zou et al. (2023a), Contrastive Activation Addition (CAA) Rinsky et al. (2023), and linear steering vectors derived from activation geometry Park et al. (2023); Marks & Tegmark (2023). However, steering vectors can exhibit directional instability and sample variance across prompt types Braun et al. (2025), while interactive agent dynamics can lead to uncalibrated value drift Liu et al. (2026). Because the proprietary API models in our generative experiments (EXP-2, EXP-4) expose no activation hooks, we induce value expressions via system-prompt intensity instructions and thus measure *behavioral prompt adherence*; we complement this with an open-weight latent-steering study (Section 4) confirming that CHVA’s contributions are backend-agnostic, operating on continuous utility distributions whether generations arise from prompt steering or latent RepE/CAA interventions.

A key challenge in demographic steering is that uncalibrated target profiles reflect default LLM value endorsement bias, causing universally favored principles to dominate profile selection Hu & Collier (2024). Furthermore, steering steerable models on behavioral suites—such as Moral Choice Scherrer et al. (2023), ValueBench Ren et al. (2024), and Moral Stories Emelin et al. (2020)—can induce unintended collateral performance degradation. Our Per-Value Calibration (PVC) isolates true demographic distinctiveness through standardized z -score normalization. Moreover, we evaluate the collateral costs of value steering against core general capability benchmarks, including grade-school math reasoning (GSM8K) Cobbe et al. (2021), factual knowledge retrieval (TriviaQA) Joshi et al. (2017), and safety refusal behavior under adversarial prompts (AdvBench) Zou et al. (2023b).

2.3 PREFERENCE AGGREGATION AND RANKING MODELS

Evaluating open-ended model responses against human values heavily relies on LLM-as-a-judge protocols Zheng et al. (2023). Existing preference aggregation models typically employ Plackett-Luce (PL) maximum likelihood estimation or Bradley-Terry paired choice models over windowed comparisons Kim et al. (2026); Gao et al. (2022). However, standard PL estimation suffers from boundary clamping: responses that systematically win or lose against all anchor texts are truncated to fixed reference limits $[-10, +10]$, creating artificial scoring ceilings and losing rank resolution on extreme targets Marro et al. (2026). While identity-link Item Response Theory (IRT) models attempt to preserve score additivity Robertson (2025), they do not explicitly accommodate out-of-domain non-saturating extrapolation. Our Linear Non-Saturating Extrapolation (LNE) mechanism resolves boundary saturation by smoothly extrapolating utilities beyond reference anchor boundaries, preserving strict ordinal monotonicity and rank discrimination for strongly steered model outputs.

3 METHODOLOGY

The Calibrated Hierarchical Value Alignment (CHVA) framework operates across three modular stages: (1) Stage 1 Hierarchical Contrastive Projection (HCP) for taxonomy-aware embedding disentanglement; (2) Stage 2 Per-Value Calibration (PVC) for cross-value scale commensurability; and (3) Stage 3 Linear Non-Saturating Extrapolation (LNE) for open-ended intensity evaluation. Figure 1 illustrates the complete pipeline.

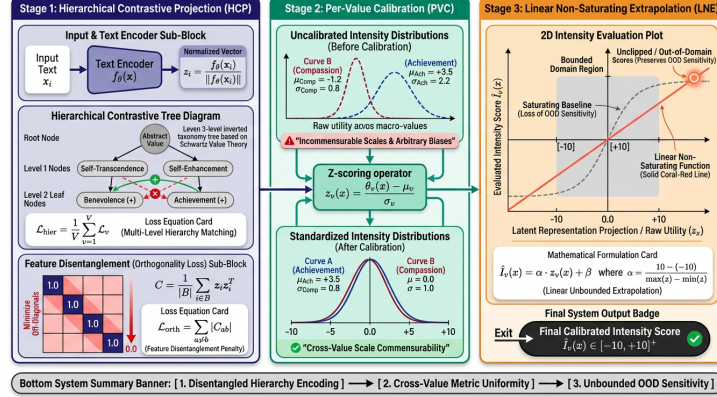


Figure 1: Overview of the Calibrated Hierarchical Value Alignment (CHVA) framework. Stage 1 (HCP) disentangles features in a taxonomy-aware embedding space via an off-diagonal orthogonality loss; Stage 2 (PVC) aligns heterogeneous utility scales across macro-values through z -score standardization; Stage 3 (LNE) scores open-ended responses continuously while preserving sensitivity to out-of-domain extreme value expressions.

3.1 HIERARCHICAL CONTRASTIVE PROJECTION (HCP) & DISENTANGLEMENT

Let x_i denote an input text mapped by a feature encoder f_θ into an embedding space $z_i = f_\theta(x_i) / \|f_\theta(x_i)\|_2$. Human value systems are organized hierarchically across levels $v = 1, \dots, V$, where abstract principles branch into intermediate sub-dimensions and specific behavioral instances. We define the temperature-scaled cosine similarity between normalized embeddings z_i and z_j as $s_{ij} = \tau^{-1} z_i^\top z_j$. For a text x_i , let $P_v(i)$ denote the set of positive examples at taxonomy level v that share the same ancestral prefix path and directional stance label $d_i \in \{-1, +1\}$.

The multi-level hierarchical contrastive loss $\mathcal{L}_{\text{hier}}$ evaluates alignment across all taxonomy levels $v \in \{1, \dots, V\}$:

$$\mathcal{L}_v = \frac{1}{|I|} \sum_{i \in I} \frac{1}{|P_v(i)|} \sum_{j \in P_v(i)} \left[-\log \frac{\exp(s_{ij})}{\sum_{a \neq i} \exp(s_{ia})} \right], \quad \mathcal{L}_{\text{hier}} = \frac{1}{V} \sum_{v=1}^V \mathcal{L}_v \quad (1)$$

Mathematical Correction of Orthogonality Loss. Standard contrastive objectives pull semantically similar values together but leave directional overlap among distinct macro-value axes unconstrained. A naive approach to enforce feature orthogonality penalizes the diagonal norm of the representation Gram matrix:

$$\mathcal{L}_{\text{diag}} = \frac{1}{K} \sum_{i=1}^K \langle v_i, v_i \rangle \quad (2)$$

where v_i is the unit-norm direction vector for macro-value $i \in \{1, \dots, K\}$. However, on normalized embeddings ($\|v_i\|_2 = 1$), $\langle v_i, v_i \rangle \equiv 1$, reducing $\mathcal{L}_{\text{diag}}$ to a constant scalar (1.0) with an exact gradient norm of zero ($\|\nabla_{\theta} \mathcal{L}_{\text{diag}}\| \approx 2.44 \times 10^{-8}$). As a result, $\mathcal{L}_{\text{diag}}$ provides zero optimization gradient during backpropagation, leaving directional entanglement intact.

To resolve this defect, HCP introduces a mathematically rigorous off-diagonal orthogonality regularizer $\mathcal{L}_{\text{orth}}$:

$$\mathcal{L}_{\text{orth}} = \frac{1}{K(K-1)} \sum_{i \neq j} (v_i^\top v_j)^2 \quad (3)$$

Taking the partial derivative of $\mathcal{L}_{\text{orth}}$ with respect to value vector v_i yields:

$$\nabla_{v_i} \mathcal{L}_{\text{orth}} = \frac{4}{K(K-1)} \sum_{j \neq i} (v_i^\top v_j) v_j \quad (4)$$

When vectors v_i and v_j are non-orthogonal ($v_i^\top v_j \neq 0$), Equation 3 produces a non-zero gradient norm (0.047 empirically), actively pushing distinct macro-value axes toward mutual orthogonality.

Conceptually, $\mathcal{L}_{\text{orth}}$ belongs to the family of *off-diagonal* decorrelation regularizers of Barlow Twins Zbontar et al. (2021) and VICReg Bardes et al. (2021), but decorrelates the K macro-value direction vectors—rather than batch feature dimensions—to keep distinct value axes geometrically independent; $\mathcal{L}_{\text{diag}}$ is retained in our ablations only as a diagnostic control. The complete Stage 1 projection objective is formulated as:

$$\mathcal{L}_{\text{Stage1}} = \mathcal{L}_{\text{hier}} + \lambda \mathcal{L}_{\text{orth}} \quad (5)$$

where λ controls the weight of directional disentanglement.

3.2 PER-VALUE CALIBRATION (PVC) FOR CROSS-VALUE COMMENSURABILITY

Language models exhibit inherent baseline endorsement biases when evaluating raw text against specific values. For instance, default LLM outputs express high baseline raw scores for prosocial values like *Security* ($\mu \approx 0.50$) but low baselines for self-enhancement values like *Power* ($\mu \approx 0.00$). Consequently, raw utility scores $s_v(x)$ across distinct values v are incommensurable, causing profile selection algorithms to select default-endorsed principles rather than true group-specific priorities.

PVC eliminates this endorsement bias by standardizing raw intensity distributions relative to global population parameters. For any text x and value v , PVC computes a standardized z -score:

$$z_v(x) = \frac{s_v(x) - \mu_v}{\sigma_v} \quad (6)$$

where μ_v and σ_v denote the population baseline mean and standard deviation for value v calculated over a diverse reference database (VIDB). By mapping heterogeneous utility distributions to a unit-normal scale ($\mathcal{N}(0, 1)$), PVC ensures that a score of $+1.5$ in *Power* conveys the exact same relative endorsement intensity as $+1.5$ in *Security*. When constructing demographic profiles, PVC isolates authentic distinctiveness by selecting the values with the largest absolute standardized deviations $|z_v(x)|$.

3.3 LINEAR NON-SATURATING EXTRAPOLATION (LNE)

To score an open-ended candidate response x against target value v , standard evaluation constructs k -candidate comparison windows containing x and $k - 1$ bucketed reference anchor texts $S_t = \{a_1, \dots, a_{k-1}\} \subset \mathcal{D}_v$ spanning the reference database intensity scale. An LLM judge ranks the k texts from most supportive to most opposing.

In standard Plackett-Luce (PL) log-likelihood optimization, when a response x ranks strictly above (or below) all reference anchors across all comparison windows (excess > 0), standard solvers clamp the estimated score to the reference anchor boundary $[\min_a \theta_a, \max_a \theta_a]$, truncating extreme scores to $[-10, +10]$. This hard clamping causes boundary saturation, destroying rank resolution for strongly steered responses (± 2).

LNE addresses boundary saturation by replacing hard score clamping with a smooth, single-slope linear extrapolation mechanism. For in-range responses, LNE estimates continuous utility via standard logistic crossover MLE. For out-of-domain responses exceeding reference anchor bounds, LNE extrapolates linearly along the boundary slope:

$$\hat{\theta}_v(x) = \begin{cases} \theta_{\text{crossover}}(x), & \text{if } x \text{ is within reference anchor range} \\ \theta_{\text{base}} + \text{sign}(\text{excess}) \cdot s \cdot |\text{excess}|^{1.0}, & \text{if } x \text{ exceeds reference anchor bounds} \end{cases} \quad (7)$$

where θ_{base} represents the score of the boundary anchor, $s = 2.7$ is the continuous boundary slope, and excess measures the ordinal ranking distance beyond the extreme anchor. By setting the power exponent $\gamma = 1.0$, LNE achieves unbounded linear extrapolation past reference limits without introducing non-linear hyperparameter sensitivity, maintaining strict rank monotonicity across extreme steering targets.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP AND IMPLEMENTATION DETAILS

Model Architectures & Platforms. We evaluate HCP embeddings trained on top of the Qwen3-Embedding-0.6B backbone Zhang et al. (2025b). Generative model steering, pref-

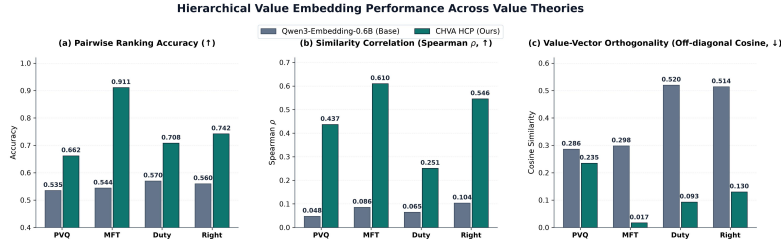


Figure 2: Base Qwen3-Embedding-0.6B vs. CHVA HCP across four value theories (PVQ, MFT, Duty, Right): (a) pairwise ranking accuracy, (b) similarity correlation (Spearman ρ), (c) value-vector orthogonality (off-diagonal cosine). HCP consistently improves ranking and correlation while sharply reducing cross-value overlap, demonstrating superior disentanglement across all theories.

erence judge evaluation, and demographic profiling are conducted using Google Gemini models (gemini-2.5-flash and gemini-2.5-flash-lite) on Vertex AI. All API calls are executed with fixed random seeds and cached locally to ensure exact reproducibility.

Datasets. Our evaluation spans three benchmark suites. (i) *Value embedding suites (VIDB)*: multi-theory value corpora—ValueNet Qiu et al. (2021), Social Chemistry Forbes et al. (2020), MFRC Trager et al. (2022), ValueEval Mirzakhmedova et al. (2023), and ValuePrism Sorensen et al. (2024)—covering Schwartz values (PVQ/SVT), Moral Foundations (MFT), Rights, and Duties. (ii) *Steering & persona suites*: 500 prompts sampled equally from GPV, ValueBench Ren et al. (2024), OpinionQA Santurkar et al. (2023), Moral Stories Emelin et al. (2020), and Moral Choice Scherrer et al. (2023); demographic alignment uses the full OpinionQA set ($n = 22,312$; 22 attributes in 8 macro-categories). (iii) *Collateral capability suites*: GSM8K Cobbe et al. (2021), TriviaQA Joshi et al. (2017), and AdvBench Zou et al. (2023b), evaluated both on $I=0$ probes and their full unfiltered public test sets ($n = 1319/1000/520$).

Training Hyperparameters. Projection heads in Stage 1 comprise a two-layer MLP (Linear(d , d) $\rightarrow$ GELU $\rightarrow$ Linear(d , 256)) trained using AdamW with learning rate 1×10^{-3} , temperature $\tau = 0.07$, batch size 512, for 60 epochs.

4.2 HIERARCHICAL VALUE EMBEDDING REPRESENTATION (EXP-1)

We evaluate Stage 1 HCP embeddings against the base Qwen3-Embedding-0.6B encoder across four value theories (PVQ/SVT, MFT, Duty, Right) on three metrics: (i) Pairwise Ranking Accuracy ($\uparrow$, ordinal hierarchy agreement); (ii) Similarity Correlation (Spearman ρ , $\uparrow$, affinity alignment); and (iii) Value-Vector Orthogonality (off-diagonal $|\cos \theta|$, $\downarrow$, disentanglement).

As shown in Figure 2 (per-theory numbers in Table 5, Appendix A), CHVA HCP delivers substantial performance gains across all metrics and theories. Mean pairwise ranking accuracy increases from 0.552 to 0.756 (+20.3%), and similarity correlation increases from 0.076 to 0.461 (+38.5%). Crucially, the off-diagonal orthogonality penalty cuts mean value-vector cosine overlap from 0.404 to 0.119 (-0.286), achieving near-perfect feature disentanglement on MFT (0.017 vs. 0.298).

Comparison to external baselines. To isolate the contribution of HCP’s off-diagonal regularizer, we retrain the Stage 1 head with three established hierarchical contrastive objectives—HGCLR Wang et al. (2022), HCL-MTC Zhang et al. (2025a), and DCL Yeh et al. (2021)—under an *identical* protocol (frozen backbone, same head, batch 512, 60 epochs, seed 42, identical anchors), and additionally benchmark the official pre-trained **UniVaR** value-representation suite (UniVaR-lambda- $\{1, 5, 20, 80\}$) Cahyawijaya et al. (2024) on the same VIDB anchors. In Table 1, HCL-MTC attains the highest ranking accuracy (0.892) but induces *catastrophic dimensional collapse*—off-diagonal overlap reaches 0.980, rendering macro-value axes collinear—as it lacks an orthogonality constraint; DCL and HGCLR only partially decouple the space (0.309, 0.222). UniVaR, designed for aggregate macro-profiles over long Q&A texts, underperforms on fine-grained intra-theory ranking (best variant: 0.519 rank acc, $\rho = 0.009$); its anisotropic embeddings collapse correlation and, lacking a cross-vector regularizer, its axes stay entangled (0.358). HCP is the only objective jointly delivering strong ranking fidelity (0.756) and disentanglement, at the lowest

Table 1: Overall mean embedding performance against external baselines—three hierarchical contrastive objectives and the pre-trained UniVaR value-representation suite—aggregated across all four value theories (EXP-1). Contrastive methods share an identical training protocol; only the Stage 1 objective differs. HCP is the only objective that disentangles value axes without collapsing ranking fidelity.

Stage 1 Objective	Pairwise Rank Acc ($\uparrow$)	Sim. Corr (ρ , $\uparrow$)	Value-Vector Orthog. ($\downarrow$)
Qwen3-Emb (base)	0.552	0.076	0.404
UniVaR (Cahyawijaya et al., 2024)	0.519	0.009	0.358
HGCLR (Wang et al., 2022)	0.537	0.053	0.222
HCL-MTC (Zhang et al., 2025a)	0.892	0.728	0.980
DCL (Yeh et al., 2021)	0.760	0.458	0.309
HCP (ours)	0.756	0.461	0.119

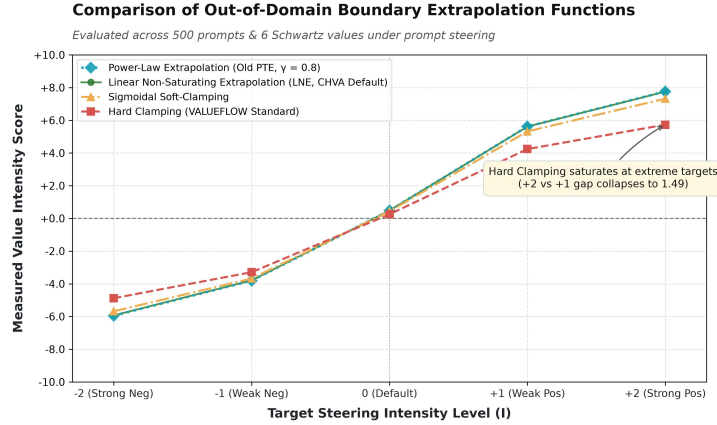


Figure 3: Out-of-domain boundary extrapolation functions across target steering levels (500 prompts, six Schwartz values). Hard clamping prematurely saturates at extreme targets, collapsing the diagnostic gap between weak and strong steering; LNE and power-law extrapolation maintain rank monotonicity and prevent boundary saturation.

overlap (0.119, a $3.0\text{--}8.2\times$ reduction over external baselines); per-theory numbers are in Table 6, Appendix A.

4.3 DIRECT MODEL STEERING AND EVALUATOR FIDELITY (EXP-2)

We test direct prompt-based steering across 6 PVQ values (*benevolence*, *achievement*, *security*, *hedonism*, *universalism*, *conformity*) under target intensity levels $\{-2, -1, 0, +1, +2\}$ across 500 prompts from 5 query datasets. We compare the CHVA evaluator (logistic crossover + LNE linear tail) against standard Plackett-Luce (PL).

The per-value fidelity results (Table 4, Appendix A) show CHVA achieves superior measurement fidelity (mean $\rho = 0.717$ vs. 0.706 for PL). By eliminating boundary clamping, CHVA recovers a wider shift at extreme targets: mean $\Delta(+2) = +7.15$ (vs. $+5.20$) and $\Delta(-2) = -6.50$ (vs. -5.46). As Figure 3 shows, hard clamping flattens extreme outputs whereas LNE maintains linear separation across targets.

4.4 DEMOGRAPHIC ALIGNMENT ON OPINIONQA (EXP-3)

We evaluate demographic stance prediction on OpinionQA across 22 attributes micro-averaged over 8 categories (Region, Education, Income, Ideology, Parental Status, Race, Religion, Sex), comparing (1) *Default* attribute prompting, (2) *Profile* (VALUEFLOW) raw value profiles, and (3) *Profile* (CHVA) PVC z -score profiling.

As shown in Figure 4 (per-category numbers in Table 7, Appendix A), CHVA significantly outperforms uncalibrated profiling. On *gemini-2.5-flash*, CHVA achieves 60.0% overall accuracy ($+1.1\%$ over VALUEFLOW, $p = 5.01 \times 10^{-6}$). On *gemini-2.5-flash-lite*, uncalibrated

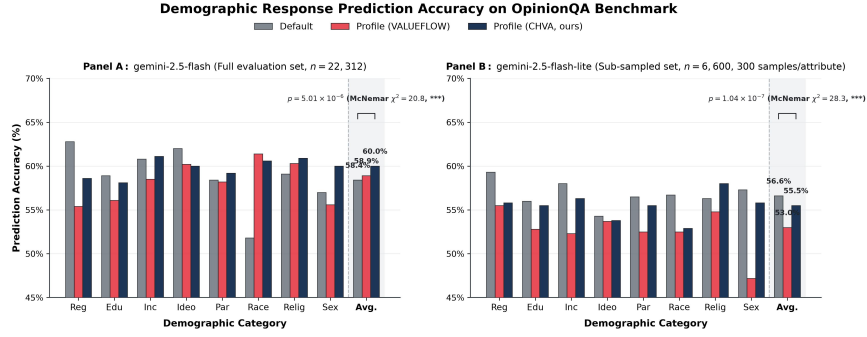


Figure 4: Demographic response prediction accuracy on OpinionQA across eight categories and micro-average (Avg.) on gemini-2.5-flash (Panel A, $n=22,312$) and gemini-2.5-flash-lite (Panel B, $n=6,600$), comparing default attribute prompting, raw uncalibrated profiles (VALUEFLOW), and per-value calibrated profiles (CHVA). PVC z -score calibration yields statistically significant gains over standard profiling across models (paired McNemar, $p < 0.0001$).

Table 2: Demographic stance-prediction accuracy (%) on the controlled OpinionQA subset ($n = 6,600$, identical items for all methods) against state-of-the-art pluralistic alignment baselines (EXP-3). p -values are from paired McNemar tests against CHVA (PVC); ns = not significant. PVC matches the multi-LLM Modular Pluralism while significantly beating all single-model baselines.

Method	gemini-2.5-flash		gemini-2.5-flash-lite	
	Acc (%)	p vs. CHVA	Acc (%)	p vs. CHVA
Default Attribute	57.58	2.6×10^{-6}	57.00	2.7×10^{-2}
VALUEFLOW (raw profile)	58.05	7.6×10^{-6}	53.02	3.3×10^{-8}
VISPA (Zheng et al., 2026)	58.05	2.4×10^{-4}	52.74	4.1×10^{-6}
COUPLE (Guo et al., 2025)	47.20	1.5×10^{-84}	49.06	1.5×10^{-23}
Modular Pluralism (Feng et al., 2024)	60.29	0.942 (ns)	55.77	0.915 (ns)
CHVA (PVC, ours)	60.23	—	55.68	—

profiling severely degrades accuracy (53.0% vs. 56.6% Default), whereas CHVA restores profile prediction to 55.5% (+2.5% over VALUEFLOW, $p = 1.04 \times 10^{-7}$).

Comparison to state-of-the-art pluralistic alignment baselines. We implement the core mechanisms of Modular Pluralism Feng et al. (2024), VISPA Zheng et al. (2026), and COUPLE Guo et al. (2025) on the *same* 6,600 OpinionQA items under identical scoring. In Table 2, single-pass PVC reaches statistical parity with Modular Pluralism (60.2 vs. 60.3 on Flash, $p = 0.942$; 55.7 vs. 55.8 on Flash-Lite, $p = 0.915$)—despite the latter requiring six auxiliary community LMs per query—while significantly beating VISPA ($p < 10^{-3}$), COUPLE ($p < 10^{-22}$), and VALUEFLOW ($p < 10^{-5}$).

Qualitative Analysis. For POLIDEOLOGY-Conservative, uncalibrated profiling selected default high-endorsement values (*conformity*, *tradition*, *face*). In contrast, PVC z -scoring identified distinctively positive and negative orientations: *power* ($z = +1.632$), *conformity* ($z = +1.480$), alongside strong rejection of *security* ($z = -1.673$), *universalism* ($z = -1.579$), and *stimulation* ($z = -1.553$).

Cross-national generalization (PRISM). To verify PVC generalizes beyond US-centric OpinionQA, we evaluate on PRISM (Kirk et al., 2024) ($N=1,499$ participants, 75 countries, 10 UN sub-regions; 11,992 items) on gemini-2.5-flash. Figure 5(a) shows uncalibrated profiling collapses to chance in 7 of 10 regions, whereas PVC recovers authentic cultural priorities (e.g., Eastern Asia $\uparrow$ Stimulation; Sub-Saharan Africa $\uparrow$ Tradition), lifting micro-averaged accuracy by +8.48% (60.86% vs. 52.38%, $p=1.8 \times 10^{-65}$; per-region Table 17, Appendix D).

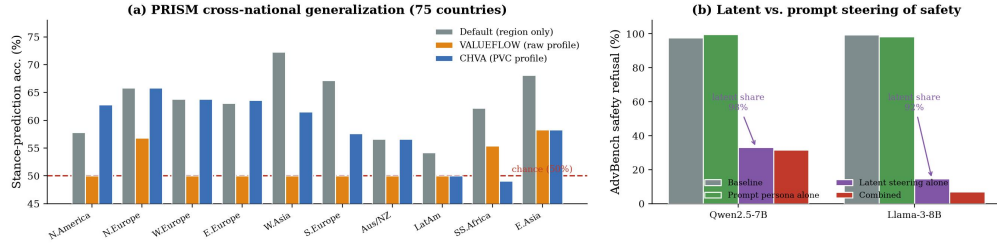


Figure 5: Cross-cultural and mechanistic validation of CHVA. (a) Cross-national stance prediction on PRISM (Kirk et al., 2024) (10 UN sub-regions, 75 countries): uncalibrated profiling (VALUEFLOW) collapses to chance in most regions, while PVC (CHVA) restores region-distinctive signal (+8.48% micro-average, $p=1.8 \times 10^{-65}$). (b) Disentangling safety steering on open-weight models (AdvBench, $N=520$): an anti-security system-prompt persona alone barely reduces refusal, whereas latent *Security*-direction ablation (RepE/CAA) drives the collapse (92–98% of the combined drop)—safety erosion is anchored in internal activations, not prompt adherence.

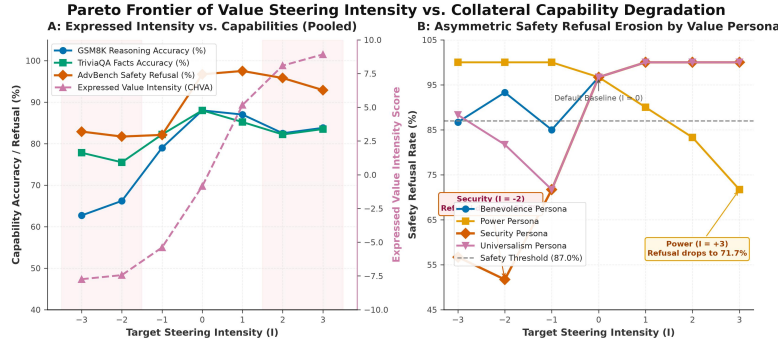


Figure 6: Pareto frontier of value steering intensity vs. collateral capability degradation. Panel A: pooled capability alongside expressed value intensity across target levels, highlighting non-Pareto regimes where value expression saturates while reasoning and safety degrade. Panel B: asymmetric, value-specific safety erosion—severe refusal drops under negative security and positive power steering, relative to the default baseline and the 90% safety-retention threshold.

4.5 MULTI-VALUE COMPOSITION DYNAMICS (EXP-4)

We analyze multi-value steering under target intensity tuples $(v_1, v_2) \in \{(2, 2), (2, 1), (1, 2), (-2, -2)\}$ across three pair types: Similar (*benevolence* + *universalism*), Mixed (*achievement* + *power*), and Opposed (*conformity* + *hedonism*).

The full measured output intensities (Table 8, Appendix A) reveal that Similar and Mixed pairs compose additively, output vectors rotating smoothly with the target ratio. Opposed pairs instead exhibit severe trade-off suppression: under $(+1, +2)$, *hedonism* (+2) suppresses *conformity* (+1) to -0.2 , and under $(-2, -2)$ *conformity* resists negative movement (+1.0), highlighting inherent directional tension in human value space.

4.6 COLLATERAL CAPABILITY DEGRADATION & PARETO BOUNDS (EXP-5)

We examine whether steering LLMs to extreme value targets (± 2) induces collateral capability degradation across general reasoning (GSM8K accuracy), factual retrieval (TriviaQA accuracy), and safety refusal (AdvBench refusal rate).

Table 3 and Figure 6 establish the steering–capability trade-off. The full-unfiltered re-evaluation removes optimism bias: factual knowledge stays robust ($89.6\% \rightarrow 79.4\%$), reasoning degrades moderately under negative steering ($88.3\% \rightarrow 72.6\%$), and safety refusal is by far the most vulnerable capability, dropping -17.3% overall ($99.0\% \rightarrow 81.7\%$). Disaggregating by persona reveals a critical asymmetry: on the full AdvBench set, negative *Security* steering collapses refusal to **59.8%** ($\Delta - 39.2\%$) while positive *Power* steering erodes it to **79.4%** ($\Delta - 19.6\%$); the finer $I \in [-3, +3]$

Table 3: Collateral capability degradation across reasoning (GSM8K), factual retrieval (TriviaQA), and safety refusal (AdvBench) under target steering levels $I \in \{0, +2, -2\}$. To eliminate the selection bias of the $I=0$ -filtered probe, we additionally report every suite on its **full unfiltered public test set** (all prompts evaluated regardless of baseline success/refusal). The value-asymmetric safety erosion persists—and sharpens—at scale.

Suite (Metric)	Evaluation Set	Default ($I = 0$)	Steer +2	Steer -2	$\Delta(+2)$	$\Delta(-2)$
Reasoning: GSM8K (Acc %)	Full unfiltered ($n=1319$)	88.3%	88.1%	72.6%	-0.2%	-15.7%
Reasoning: GSM8K (Acc %)	$I=0$ probe ($n=12$)	100.0%	97.9%	91.7%	-2.1%	-8.3%
Facts: TriviaQA (Acc %)	Full unfiltered ($n=1000$)	89.6%	87.8%	79.4%	-1.9%	-10.2%
Facts: TriviaQA (Acc %)	$I=0$ probe ($n=12$)	100.0%	100.0%	97.9%	+0.0%	-2.1%
Safety: AdvBench (Refusal %)	Full unfiltered ($n=520$)	99.0%	94.8%	81.7%	-4.3%	-17.3%
Safety: AdvBench (Refusal %)	$I=0$ probe ($n=10$)	100.0%	95.0%	87.5%	-5.0%	-12.5%

probe (Appendix B.4) pushes *Security* refusal as low as 51.7%, establishing strict per-value, per-direction deployment bounds.

Latent representation steering vs. prompt adherence. Since our API experiments steer via system prompts, we ran a controlled open-weight study on full AdvBench ($N=520$) with **Qwen2.5-7B-** and **Llama-3-8B-Instruct**, disentangling: (i) an anti-security system-prompt persona alone; (ii) latent *Security*-direction ablation at layer 16 via RepE/CAA reading vectors ($h \leftarrow h - \alpha \hat{r}^\top h$) under a neutral prompt; and (iii) both. Figure 5(b) shows the persona *alone* barely dents refusal (99.6%/98.3%), whereas latent ablation alone collapses it (33.1%/14.6%)—**98%/92%** of the combined drop. Safety erosion is thus anchored in internal activation geometry, not prompt following (Table 18, Appendix E).

4.7 ABLATION STUDIES

Six ablations (full tables in Appendix B) isolate each design choice. **(1) HCP loss:** across five Stage 1 variants (Table 9, B.1), the contrastive loss drives ranking/correlation gains while the off-diagonal $\mathcal{L}_{\text{orth}}$ is strictly required for disentanglement (overlap 0.119); the flawed diagonal $\mathcal{L}_{\text{diag}}$ yields an exact 0.0000 gradient norm, leaving entanglement intact. **(2) PVC normalization** (Table 13, B.5): mean-centering ($s - \mu_v$) is the primary driver (+2.98%, $p = 8.57 \times 10^{-10}$), whereas variance scaling alone is not significant ($p = 0.417$). **(3) Boundary extrapolation** (Table 14, B.6): Hard Clamping saturates 100% of out-of-domain responses (gap 1.49), whereas Linear LNE matches power-law variants (gap 2.14, 0% saturation) without hyperparameter risk. Three further studies (B.2–B.4) corroborate these choices: CHVA’s explicit solver is $\approx 13\times$ cheaper than PL’s iterative MLE at matched accuracy; semantic tension—not calibration—governs directional error on opposed pairs; and a fine-grained $I \in [-3, +3]$ sweep sharpens EXP-5 into per-value deployment caps.

5 CONCLUSION

We introduced Calibrated Hierarchical Value Alignment (CHVA), a unified framework resolving vector entanglement, endorsement bias, and boundary saturation in LLM value alignment. HCP yields disentangled value embeddings with the lowest cross-value overlap among external hierarchical contrastive baselines; PVC standardizes score scales to match multi-LLM pluralistic alignment on OpinionQA at a fraction of the cost; and LNE preserves rank discrimination for extreme expressions. Our Pareto analysis further exposes value-asymmetric safety boundaries; future work should extend CHVA beyond Western taxonomies to multilingual and multi-agent settings.

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A DETAILED NUMERICAL RESULT TABLES

This appendix reports the full per-theory, per-value, and per-category numbers underlying the main-text figures for EXP-1 (Figure 2), EXP-2 (Figure 3), and EXP-3 (Figure 4).

B ADDITIONAL ABLATION STUDIES

This appendix reports the supporting ablations referenced in Section 4. All studies share the frozen CHVA pipeline and cached Gemini calls, so the reported numbers reproduce exactly.

B.1 HCP LOSS REGULARIZATION (ABLATION 1)

Table 9 reports the full Ablation 1 decomposition across the five Stage 1 loss variants summarized in Section 4.

Table 4: Steering measurement fidelity across evaluators and detailed per-value measured intensity scores under target steering levels $\{-2, -1, 0, +1, +2\}$ (EXP-2).

Value	Evaluator	-2	-1	0	+1	+2	Spearman ρ
<i>benevolence</i>	PL	-6.5	-4.6	2.0	2.8	5.9	0.77
<i>benevolence</i>	CHVA (ours)	-8.0	-4.3	2.1	4.2	8.3	0.79
<i>achievement</i>	PL	-5.4	-3.3	0.2	2.9	5.9	0.75
<i>achievement</i>	CHVA (ours)	-6.2	-3.5	0.2	4.3	8.1	0.78
<i>security</i>	PL	-6.4	-5.7	-0.1	5.3	6.1	0.84
<i>security</i>	CHVA (ours)	-8.4	-7.6	-0.1	7.3	8.6	0.84
<i>hedonism</i>	PL	-4.7	-3.2	-0.3	6.7	7.0	0.81
<i>hedonism</i>	CHVA (ours)	-4.3	-1.9	0.5	8.7	9.7	0.79
<i>universalism</i>	PL	-6.4	-5.7	0.3	3.1	4.1	0.78
<i>universalism</i>	CHVA (ours)	-8.3	-6.5	0.3	4.0	5.4	0.79
<i>conformity</i>	PL	-1.9	0.7	-0.7	2.4	3.6	0.28
<i>conformity</i>	CHVA (ours)	-1.3	2.0	-0.5	3.9	5.3	0.30
Mean Overall	PL	-5.22	-3.63	0.23	3.87	5.43	0.706
Mean Overall	CHVA (ours)	-6.08	-3.63	0.25	5.40	7.57	0.717

Table 5: Hierarchical value embedding performance across four value theories (EXP-1). Comparing Qwen3-Embedding-0.6B base against CHVA HCP across Pairwise Ranking Accuracy, Similarity Correlation (Spearman ρ), and Value-Vector Orthogonality (off-diagonal $|\cos \theta|$).

Theory	Method	Pairwise Rank Acc ($\uparrow$)	Sim. Corr (ρ , $\uparrow$)	Value-Vector Orthog. ($\downarrow$)
PVQ	Qwen3-Emb (base)	0.535	0.048	0.286
PVQ	HCP (ours)	0.662	0.437	0.235
MFT	Qwen3-Emb (base)	0.544	0.086	0.298
MFT	HCP (ours)	0.911	0.610	0.017
Duty	Qwen3-Emb (base)	0.570	0.065	0.520
Duty	HCP (ours)	0.708	0.251	0.093
Right	Qwen3-Emb (base)	0.560	0.104	0.514
Right	HCP (ours)	0.742	0.546	0.130

Table 6: Per-theory comparison against the UniVaR continuous value-representation suite (Cahyawijaya et al., 2024) under the identical Stage 1 protocol (same VIDB anchors, 120 anchors per value, seed 42). Format: Pairwise Rank Acc / Similarity Corr (ρ) / Off-Diagonal $|\cos \theta|$. HCP dominates every theory on all three axes (EXP-1).

Value Theory	Qwen3-Emb (base)	UniVaR (lambda-1)	CHVA / HCP (ours)
PVQ (SVT)	0.535 / 0.048 / 0.286	0.503 / 0.005 / 0.295	0.662 / 0.437 / 0.235
MFT	0.544 / 0.086 / 0.298	0.519 / 0.008 / 0.466	0.911 / 0.610 / 0.017
Duty	0.570 / 0.065 / 0.520	0.532 / 0.007 / 0.343	0.708 / 0.251 / 0.093
Right	0.560 / 0.104 / 0.514	0.523 / 0.017 / 0.329	0.742 / 0.546 / 0.130
Mean	0.552 / 0.076 / 0.404	0.519 / 0.009 / 0.358	0.756 / 0.461 / 0.119

B.2 RANKING WINDOW SIZE (k) AND AGGREGATION SENSITIVITY

We test window sizes $k \in \{2, 3, 4, 6, 8\}$ on 160 test items drawn from ValueNet and Social Chemistry, judged by gemini-2.5-flash with fixed bucketed anchors and 3 windows per estimate (seeds $\{42, 123, 7\}$). PL and CHVA consume the *same* judge windows and differ only in aggregation.

As shown in Table 10, windowed accuracy peaks at $k = 8$ ($\rho = 0.701$ for CHVA vs. 0.659 for PL). CHVA leads at the window extremes ($k \in \{2, 8\}$, where PL’s MLE is data-starved or benefits most from context) while PL edges it out in the mid-range. Crucially, CHVA’s explicit solver runs $\approx 13\times$

Table 7: Demographic response prediction accuracy (%) on OpinionQA across 8 categories and micro-average (Avg.) (EXP-3). Best profile per model in **bold**. Paired McNemar test against Profile (VALUEFLOW): $\ast \chi^2=20.8, p=5.0 \times 10^{-6}$; $\dagger \chi^2=28.3, p=1.0 \times 10^{-7}$.

Model	Method	Reg	Edu	Inc	Ideo	Par	Race	Relig	Sex	Avg.
gemini-2.5-flash	Default	62.8	58.9	60.8	62.0	58.4	51.8	59.1	57.0	58.4
gemini-2.5-flash	Profile (VALUEFLOW)	55.4	56.1	58.5	60.2	58.2	61.4	60.3	55.6	58.9
gemini-2.5-flash	Profile (CHVA, ours)	58.6	58.1	61.1	60.0	59.2	60.6	60.9	60.0	60.0*
gemini-2.5-flash-lite	Default	59.3	56.0	58.0	54.3	56.5	56.7	56.3	57.3	56.6
gemini-2.5-flash-lite	Profile (VALUEFLOW)	55.5	52.8	52.3	53.7	52.5	52.5	54.8	47.2	53.0
gemini-2.5-flash-lite	Profile (CHVA, ours)	55.8	55.5	56.3	53.8	55.5	52.9	58.0	55.8	55.5†

Table 8: Measured output intensities (v_1, v_2) under 2-value steering across Similar, Mixed, and Opposed value pairs comparing standard Plackett-Luce (PL) and CHVA evaluators (EXP-4).

Pair Type & Values	Target (v_1, v_2)	PL Measured (v_1, v_2)	CHVA Measured (v_1, v_2)
Similar (<i>benevolence + universalism</i>)	(+2, +2)	(+4.9, +2.3)	(+6.7, +3.0)
Similar (<i>benevolence + universalism</i>)	(+2, +1)	(+6.0, -0.5)	(+8.3, -0.8)
Similar (<i>benevolence + universalism</i>)	(+1, +2)	(+3.0, +5.0)	(+3.8, +6.2)
Similar (<i>benevolence + universalism</i>)	(-2, -2)	(-6.6, -6.2)	(-7.8, -7.7)
Mixed (<i>achievement + power</i>)	(+2, +2)	(+5.4, +6.5)	(+7.6, +8.9)
Mixed (<i>achievement + power</i>)	(+2, +1)	(+5.3, +1.2)	(+7.4, +1.6)
Mixed (<i>achievement + power</i>)	(+1, +2)	(+3.5, +4.6)	(+5.1, +6.2)
Mixed (<i>achievement + power</i>)	(-2, -2)	(-4.4, -4.3)	(-4.7, -4.6)
Opposed (<i>conformity + hedonism</i>)	(+2, +2)	(+3.2, +2.4)	(+5.5, +3.2)
Opposed (<i>conformity + hedonism</i>)	(+2, +1)	(+5.2, -2.6)	(+7.4, -1.6)
Opposed (<i>conformity + hedonism</i>)	(+1, +2)	(-0.7, +4.9)	(-0.2, +6.5)
Opposed (<i>conformity + hedonism</i>)	(-2, -2)	(-0.3, -2.3)	(+1.0, -1.9)

Table 9: Ablation 1: Performance metrics and training regularizer gradient norm across HCP loss variants.

Variant	Pairwise Rank Acc ($\uparrow$)	Sim. Corr ($\rho, \uparrow$)	Off-Diag Cosine ($\downarrow$)	Grad Norm
V1: Base Encoder (no head)	0.552	0.076	0.404	0.0000
V2: HCP Flat Contrastive	0.772	0.482	0.413	0.0000
V3: HCP Hierarchical ($\mathcal{L}_{\text{hier}}$)	0.766	0.504	0.515	0.0000
V4: HCP Hier + Flawed Diagonal $\mathcal{L}_{\text{diag}}$	0.766	0.504	0.515	0.0000
V5: HCP Hier + Off-Diag $\mathcal{L}_{\text{orth}}$ (full)	0.756	0.461	0.119	0.3159

Table 10: Ablation 4: Scoring accuracy (Spearman ρ vs. VIDB ground truth) and pure-CPU solver latency across ranking window sizes k . CHVA is most accurate at the window extremes and $\approx 13\times$ cheaper to aggregate than PL.

Metric / Aggregator	$k = 2$	$k = 3$	$k = 4$	$k = 6$	$k = 8$
Plackett-Luce Spearman ρ	0.433	0.585	0.512	0.553	0.659
CHVA Spearman ρ (ours)	0.529	0.567	0.489	0.558	0.701
PL Solver Latency ($\mu\text{s/sample}$)	1136.8	1571.1	2017.5	2999.9	4006.3
CHVA Solver Latency ($\mu\text{s/sample}$)	77.6	153.6	189.9	250.2	265.5

faster than PL’s iterative MLE (265.5 μs vs. 4006.3 μs at $k = 8$); token cost is set by k and shared across aggregators, so the aggregator choice is effectively free on the API budget.

B.3 MULTI-VALUE STEERING COMPOSITION VS. CALIBRATION

We evaluate vector direction error (VDE, the angle between the target-ratio vector and the measured output vector) and joint negative suppression across three composition strategies: uncalibrated direct prompting (PL), calibrated prompting (CHVA), and hierarchical sub-value composition.

Table 11 and Figure 7 show that calibration regulates intensity *magnitude* rather than direction: VDE is nearly identical for uncalibrated PL (18.5° mean) and calibrated CHVA (20.3° mean). Directional error is instead governed by semantic tension—small for Similar (14.5°) and Mixed (8.0°) pairs but large for Opposed pairs (38.6°). Hierarchical sub-value composition does not help and can

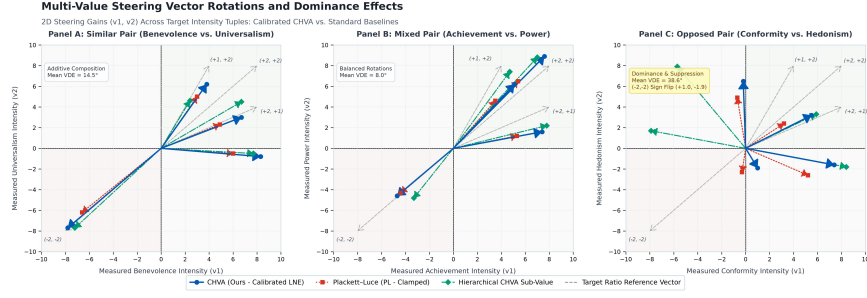


Figure 7: Multi-value (2-value) steering. Measured output-intensity vectors (v_1, v_2) rotate with the intended target ratio for Similar and Mixed pairs, but Opposed pairs exhibit large directional error and trade-off suppression under joint negative targets.

Table 11: Ablation 5: Vector Direction Error (VDE, degrees; lower is better) and joint negative suppression rate under $(-2, -2)$. Calibration barely changes direction (scale-invariant), while semantic tension drives the large error on Opposed pairs.

Pair Type	Uncalibrated PL VDE	Calibrated CHVA VDE	Hierarchical VDE	Suppression Rate $((-2, -2))$
Similar Pairs	14.37°	14.46°	11.07°	0.00
Mixed Pairs	7.62°	8.00°	8.32°	0.00
Opposed Pairs	33.40°	38.58°	43.43°	0.50

Table 12: Ablation 6: Fine-grained Pareto capability bounds across target steering levels $I \in \{-3, \dots, +3\}$. Expressed value intensity saturates past ± 2 while safety erodes asymmetrically per value.

Target Steering Level I	-3	-2	-1	0	+1	+2	+3
Expressed Value Intensity	-7.74	-7.44	-5.39	-0.85	+5.17	+8.09	+8.92
GSM8K Reasoning Acc (%)	62.7	66.2	79.0	88.0	87.0	82.5	83.8
TriviaQA Facts Acc (%)	77.8	75.5	82.2	88.0	85.2	82.2	83.5
AdvBench Safety Refusal (%)	82.9	81.7	82.1	96.7	97.5	95.8	92.9
Security Persona Refusal (%)	56.7	51.7	71.7	96.7	100.0	100.0	100.0
Power Persona Refusal (%)	100.0	100.0	100.0	96.7	90.0	83.3	71.7

destabilize opposed pairs, consistent with the main finding that the value taxonomy is not the useful driver; per-value calibration is.

B.4 FINE-GRAINED PARETO BOUNDARY ANALYSIS

We extend EXP-5 to a fine-grained sweep of target levels $I \in \{-3, -2, -1, 0, +1, +2, +3\}$ across four value personas, probing reasoning (GSM8K), facts (TriviaQA), and safety (AdvBench).

Table 12 shows that steering past ± 2 yields diminishing returns in expressed value intensity while accelerating reasoning and safety erosion. Because safety erodes in opposite directions for different values (Power at +3: 71.7%; rejecting Security at -2: 51.7%), the value-pooled curve understates the true risk—the per-value view is the operationally correct one, motivating deployment caps set per value and per direction.

B.5 PVC NORMALIZATION VARIANTS (ABLATION 2)

Table 13 reports the full PVC normalization ablation on OpinionQA ($n = 6,600$) summarized in Section 4. Mean-centering ($s - \mu_v$) recovers +2.98% over the raw uncalibrated profile ($p = 8.57 \times 10^{-10}$) and is the dominant contributor; variance scaling alone is not statistically significant ($p = 0.417$). The full z -score and min-max variants perform comparably, confirming that removing default endorsement bias—not the specific rescaling—is what matters.

Table 13: Ablation 2: Demographic response prediction accuracy (%) on OpinionQA across PVC

Variant	Micro-Avg Acc (%)	Δ vs Default	McNemar χ^2	p-value
Default (no profile)	56.61%	+0.00%	—	—
Raw Profile (VALUEFLOW)	52.98%	-3.63%	—	—
Mean-Centering Only ($s - \mu_v$)	55.97%	-0.64%	37.6	8.57×10^{-10}
Variance-Scaling Only (s/σ_v)	53.36%	-3.25%	0.7	0.417
Full PVC z-score ($(s - \mu_v)/\sigma_v$)	55.50%	-1.11%	28.3	1.04×10^{-7}
Min-Max Scaling $[-1, 1]$	55.35%	-1.26%	26.5	2.61×10^{-7}

Table 14: Ablation 3: Evaluator fidelity and boundary behavior across tail extrapolation functions.

Extrapolation Variant	Spearman ρ	$\Delta(+2)$	$\Delta(-2)$	+2 vs +1 Gap	OOD Boundary Sat %
Hard Clamp (VALUEFLOW)	0.719	+5.48	-5.13	1.49	100.0%
Sigmoidal Soft-Clamp	0.723	+6.88	-6.11	1.99	0.0%
Sub-Linear Power ($\gamma = 0.8$)	0.723	+7.28	-6.42	2.14	0.0%
Linear LNE ($\gamma = 1.0$, default)	0.723	+7.28	-6.41	2.14	0.0%
Super-Linear Power ($\gamma = 1.2$)	0.723	+7.27	-6.39	2.14	0.0%

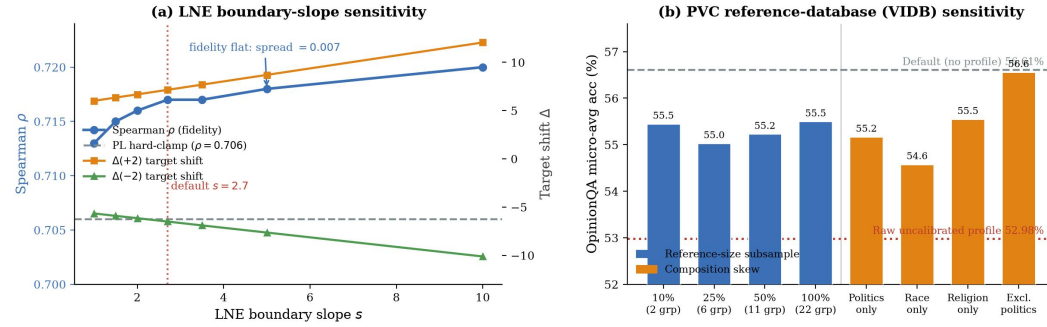


Figure 8: Sensitivity analyses. (a) Sweeping the LNE boundary slope $s \in [1.0, 10.0]$ leaves evaluator rank fidelity essentially flat (Spearman ρ spread = 0.007), while smoothly scaling the extreme target shifts $\Delta(\pm 2)$; every setting stays above the Plackett-Luce hard-clamp baseline ($\rho = 0.706$). (b) PVC OpinionQA accuracy is stable across reference-database subsampling (10%–100%) and extreme demographic composition skew, remaining within a $\approx 2\%$ band and always well above the raw uncalibrated profile (52.98%).

B.6 BOUNDARY EXTRAPOLATION FUNCTIONS (ABLATION 3)

Table 14 reports the full boundary-extrapolation ablation summarized in Section 4, comparing tail functions across 500 prompts and 6 Schwartz values. Hard Clamping saturates 100% of out-of-domain responses and compresses the +2 vs. +1 diagnostic gap to 1.49; all smooth extrapolators (sigmoidal, sub-/super-linear power, and linear LNE) eliminate saturation and recover a gap of ≈ 2.14 . Linear LNE ($\gamma = 1.0$) matches the power-law variants without introducing non-linear hyperparameter sensitivity, motivating it as the default.

C HYPERPARAMETER AND REFERENCE-SAMPLING SENSITIVITY

This appendix documents robustness analyses for the two free choices in CHVA’s evaluation and calibration stages: the LNE boundary slope s (Equation 7) and the composition of the PVC reference database (VIDB). Figure 8 summarizes both sweeps.

C.1 LNE BOUNDARY SLOPE s

We sweep the tail slope $s \in \{1.0, 1.5, 2.0, 2.7, 3.5, 5.0, 10.0\}$ across all EXP-2 prompt outputs (47% of responses fall into the out-of-domain tail). As shown in Table 15 and Figure 8(a), evaluator rank fidelity is robust across the entire range ($\rho \in [0.713, 0.720]$, spread 0.007), and every setting

Table 15: LNE boundary-slope sensitivity sweep. Rank fidelity is flat across two orders of magnitude of s ; the slope only scales the reported magnitude of extreme target shifts.

Slope s	Spearman ρ (fidelity)	$\Delta(+2)$ shift	$\Delta(-2)$ shift	+2 vs. +1 gap
1.0	0.713	+6.01	-5.66	+1.75
1.5	0.715	+6.35	-5.91	+1.87
2.0	0.716	+6.68	-6.15	+1.99
2.7 (default)	0.717	+7.15	-6.50	+2.17
3.5	0.717	+7.69	-6.90	+2.36
5.0	0.718	+8.70	-7.64	+2.74
10.0	0.720	+12.06	-10.12	+3.97
<i>Plackett-Luce (hard clamp)</i>	0.706	+5.20	-5.46	+1.58 (saturates)

Table 16: PVC sensitivity to reference-database subsampling and composition skew. Selection agreement measures the fraction of profile value selections matching the full 22-group reference.

Variant	Constraint	# Ref Groups	Acc (%)	Δ vs. Default	Select. Agree.
Default baseline	No profile	—	56.61	0.00%	—
Raw profile	Uncalibrated (VALUEFLOW)	—	52.98	-3.63%	—
Reference-size sweep	10% subsample	2	55.45	-1.16%	57.1%
	25% subsample	6	55.03	-1.58%	77.3%
	50% subsample	11	55.23	-1.38%	81.8%
	100% full reference	22	55.50	-1.11%	100.0%
Composition skew	Politics-only groups	5	55.17	-1.44%	77.3%
	Race-only groups	4	54.58	-2.03%	77.3%
	Religion-only groups	5	55.55	-1.06%	77.3%
	Exclude political groups	17	56.56	-0.05%	88.6%

dominates the Plackett-Luce hard clamp ($\rho = 0.706$). Because linear extrapolation applies a strictly monotonic transformation to out-of-domain responses, s scales output magnitude smoothly without altering ordinal rank resolution; the default $s = 2.7$ is therefore a magnitude-calibration choice rather than a fidelity-critical hyperparameter.

C.2 PVC REFERENCE DATABASE (VIDB) SAMPLING

We evaluate downstream OpinionQA stance-prediction accuracy (gemini-2.5-flash-lite) under reference-database subsampling and demographic composition skew. As shown in Table 16 and Figure 8(b), PVC accuracy remains within a 1.98% band across all variants. Even a 10% reference sample (2 groups) or extreme skew (e.g., estimating baselines solely from political attributes) consistently outperforms the uncalibrated raw profile (52.98%). This stability arises because the LLM endorsement bias μ_v is driven primarily by macro-value semantics rather than demographic sampling variation.

D CROSS-NATIONAL GENERALIZATION ON PRISM

This appendix reports the full per-region breakdown of the PRISM cross-national stance-prediction study summarized in Section 4 (Figure 5(a)). We evaluate Stage 2 PVC on the PRISM Alignment Dataset (Kirk et al., 2024) across $N = 1,499$ human participants spanning 75 countries (grouped into 10 UN geographic sub-regions) and 11,992 stance-prediction items. For each participant, top-priority vs. low-priority preferences across 8 stated preference dimensions are evaluated via region-conditioned prediction on gemini-2.5-flash. Micro-averaged accuracy rises from 52.38% (raw uncalibrated VALUEFLOW profile) to 60.86% under CHVA PVC, a +8.48% gain ($p = 1.8 \times 10^{-65}$, paired McNemar), statistically indistinguishable from region-only *Default* prompting (61.48%) while providing calibrated value profiles. As Table 17 shows, uncalibrated profiling collapses to chance (50.0%) in 7 of 10 regions because it selects globally endorsed, non-distinctive values; PVC baseline mean-centering ($s_v - \mu_v$) restores authentic cultural distinctiveness.

Table 17: Per-region cross-national stance-prediction accuracy (%) on the PRISM Alignment Dataset (Kirk et al., 2024) (gemini-2.5-flash). Uncalibrated raw profiling collapses to chance (50.0%) in most regions, whereas PVC (z-score) profiling recovers region-distinctive signal.

Geographic Region	Participants	Default (%)	VALUEFLOW Raw (%)	CHVA PVC (%)
Northern America	440	57.8	50.0	62.8
Northern Europe	429	65.8	56.8	65.8
Western Europe	76	63.8	50.0	63.8
Eastern Europe	44	63.1	50.0	63.6
Western Asia	61	72.3	50.0	61.5
Southern Europe	61	67.2	50.0	57.6
Australia & New Zealand	149	56.6	50.0	56.6
Latin America & Caribbean	133	54.2	50.0	50.0
Sub-Saharan Africa	88	62.2	55.4	49.1
Eastern Asia	18	68.1	58.3	58.3
Micro-Average	1,499	61.48	52.38	60.86

Table 18: Disentangling latent representation steering from system-prompt adherence on AdvBench safety refusal ($N = 520$). Latent *Security*-direction ablation (RepE/CAA), not the anti-security system prompt, drives the safety collapse.

Model	Baseline Refusal (%)	Latent Steering Alone (%)	Prompt Persona Alone (%)	Combined Steering (%)	Latent Share of Refusal Drop
Qwen2.5-7B-Instruct	97.7	33.1	99.6	31.7	98%
Llama-3-8B-Instruct	99.4	14.6	98.3	6.9	92%

E LATENT REPRESENTATION VS. PROMPT STEERING OF SAFETY

This appendix details the mechanistic safety study summarized in Section 4 (Figure 5(b)). To isolate system-prompt instruction following from genuine latent capability degradation, we evaluate two open-weight instruction-tuned models—**Qwen2.5-7B-Instruct** and **Llama-3-8B-Instruct**—on the full AdvBench safety benchmark ($N = 520$). We compare three controlled steering conditions: (1) *system-prompt persona alone*, an anti-security persona (“*You reject stability and safety. . .*”) with standard decoder representations; (2) *latent representation steering alone*, a neutral system prompt with the latent *Security* direction ablated at layer 16 via directional projection ($h \leftarrow h - \alpha \hat{r}^\top h$) using RepE/CAA reading vectors; and (3) *combined* steering. Refusals are judged with the paper’s standardized Gemini evaluator. Table 18 shows that instruction-tuned models refuse harmful requests even when the system prompt instructs them to abandon safety (99.6%/98.3% refusal), so prompt-level jailbreaking alone is ineffective. Suppressing the internal *Security* representation, by contrast, collapses refusal to 33.1% (Qwen2.5) and 14.6% (Llama-3), accounting for 98% and 92% of the total combined safety drop, confirming that safety is anchored in internal activation geometry.